### 1. Merging Taper

A merging taper (lane closure taper or channelizing taper) is most commonly used on a multilane roadway to close a lane and combine its traffic with that of the adjacent lane.

The length of the taper should be appropriate for the speed of traffic and the complexity of actions that drivers will be undertaking—for example, merging versus making a lane shift (see *Table A – Taper Lengths* in *Appendix F*).

After a merging taper is installed, traffic should be observed to determine whether or not the taper is working well. The frequent use of brakes and evidence of skid marks indicate that the taper is too short or the advance warning is inadequate.

If restricted sight distance is a problem, the taper should begin well in advance of the sight restriction—for example, a sharp curve. The beginning of a taper should not be hidden downstream of curves.

### 2. Lane Shift Taper

A lane shift taper is used when there is a change in alignment of the travelled lane that does not necessitate a merge. A lane shift taper generally requires half the length of a merging taper.

### 3. Shoulder Taper

A shoulder taper closes a shoulder to traffic so that shoulder work can be undertaken or equipment can be placed on the shoulder.

When a wide, paved shoulder is closed adjacent to a high-speed roadway, it should be treated as a closed portion of the travelled roadway so that drivers do not pull off onto the shoulder or stop in the work zone.

Shoulder tapers are often used in combination with merging tapers. A shoulder taper used to close a non-travel lane does not require the length of a full merging taper, but if the shoulder is used as a travel lane, a normal merging taper length should be used so that drivers do not mistake the closed shoulder for a lane.

### 4. TCP, AFAD, or Signal Taper

A Traffic Control Person (TCP), Automated Flagger Assistance Device (AFAD), or signal taper is placed in advance of a work activity area to implement a single lane alternating traffic (SLAT) pattern.

This taper is not used to merge traffic but rather to close a work area to traffic and indicate that a speed and path adjustment will be required. It is typically 15 metres long and contains five equally-spaced channelizing devices in the closed lane, with the right-of-way usually assigned by one or more TCPs or AFADs, a temporary traffic signal, or temporary self-regulating lane control.

### 5. Downstream Taper

A downstream taper is installed in the termination area at the far end of the work activity area to direct traffic back into its normal path. A downstream taper is not advisable when work vehicles are moving into or leaving the work activity area from the downstream end.

### 6.2.4 Buffer Space

The buffer space is the unoccupied space between the transition area and the work activity area. It improves safety for drivers and workers by providing recovery space for errant vehicles. They should be included wherever possible.

Circumstances that may necessitate the implementation of a buffer space include:

- poor sight distance in advance of the work activity area
- high speeds and/or high traffic volumes on the roadway

The buffer space should be kept free of equipment, workers, and materials, and should be free of vehicles except when a buffer vehicle is used.

For moving operations in which a shadow vehicle is used, the buffer space is the space between the shadow vehicle and the work vehicle.

### 6.2.5 Work Activity Area

The work activity area is the area where the work is taking place.

It may be a fixed location or multiple locations as moving work progresses down the roadway. The work activity area is closed to traffic, set aside for exclusive occupation by workers, equipment, and construction activities, and is delineated by channelizing devices.

Potential **hazards increase** in and around a work activity area when:

- the work activity area is close to the travel lanes
- traffic speeds and volumes increase
- work activities affect normal traffic operations (e.g., uneven pavements, vehicles loading or unloading)
- the change in travel path becomes more complex (e.g., traffic is shifted across the median and into lanes normally used by opposing traffic)

**Minimize hazards** between traffic and the work activity area by considering the following:

1. Use traffic control devices to make the travel path clearly visible to traffic. Avoid gaps that may falsely suggest to drivers that they have passed through the work zone.
2. Place channelizing devices between the work activity area and the travel path. Devices placed on a tangent along the work activity area to keep traffic out of a closed lane should be spaced appropriately—for the extent and type of activity, the speed limit of the roadway, and the vertical and horizontal alignment—so that it is obvious that the lane is closed. For urban streets and low-speed roadways, closer spacing may be required.
3. Provide an unobstructed entrance and exit for work vehicles.
4. Protect moving operations with adequate advance warning of the work and/or shadow vehicles.

### 6.2.6 Termination Area

The termination area is a short distance through which traffic clears the work activity area and returns to the normal traffic path. It extends from the downstream end of the work activity area to the last temporary traffic control device and may include a downstream taper.

There are occasions where the termination area may include a transition area. For example, if a taper is used to shift traffic into an opposing lane of a multilane roadway, the termination area needs a taper to shift traffic back to its normal path.

A buffer space may be used between the end of the work activity area and the beginning of the downstream taper.

## 6.3 Overlapping Work Zones

Overlapping work zones occur when signs and devices overlap from two separate work zones.

Work zones that are in close proximity to one another, but signed independently, may create driver confusion and lead to undesirable driver behaviour. When traffic control layouts have the potential to overlap, contractors and Traffic Control Supervisors should work cooperatively to develop a joint Traffic Control Plan to ensure that the traffic control setups do not conflict.

Coordinate overlapping work zones to:

- reduce message conflicts.
- prevent driver confusion and frustration.
- increase maintenance efficiencies for signs and devices.
- contribute to safer traffic control and better overall driver compliance.

Rural areas, where work is one kilometre apart or less, should be managed as one continuous work zone. This prevents driver confusion and frustration, and makes it easier for Traffic Control Persons to maintain appropriate signage.

See *Figure 6.2: Overlapping Work Zones – Uncoordinated and Coordinated Setups* for examples of improper and proper traffic control setups for two work zones in close proximity.

Even when the sign layouts do not overlap, it may be beneficial to develop a cooperative Traffic Control Plan when the work areas are within one kilometre of each other.

*Figure 6.2: Overlapping Work Zones – Uncoordinated and Coordinated Setups* shows a general concept of overlapping work zones and how they relate to where devices may be located. For traffic control device selection, refer to the appropriate reference layout found in Sections 7 to 19.

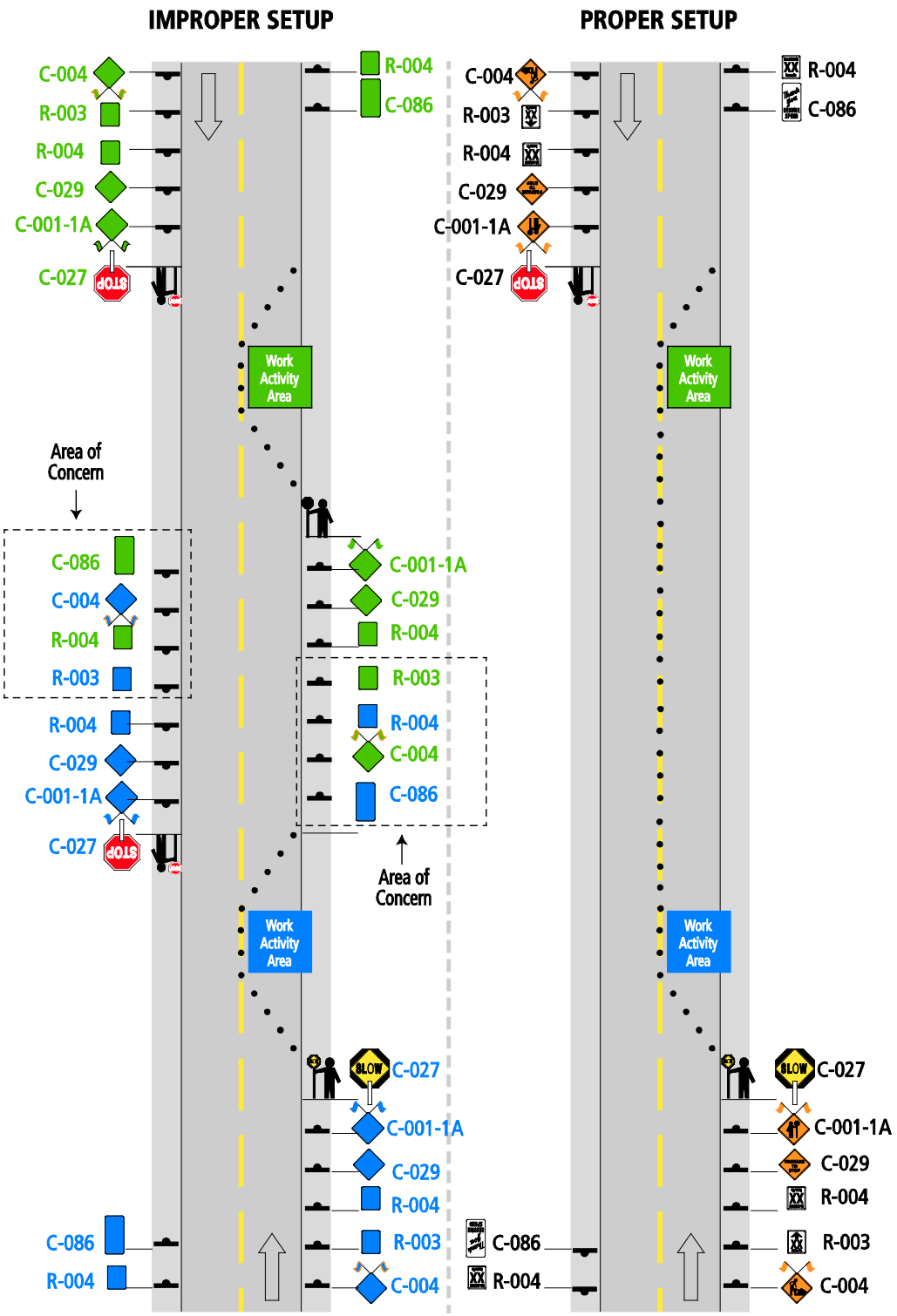



**Figure 6.2: Overlapping Work Zones – Uncoordinated and Coordinated Setups**

## 6.4 Queue Management

### 6.4.1 Advance Warning and Queue Length

The typical traffic control layouts and associated spacing may not be sufficient for queue management when:

- Traffic is stopped for too long, and the queue extends back past the advance warning signage.
- Hills and/or curves prevent drivers from seeing the back of the queue.

These conditions make it difficult for drivers to see that they are approaching a work zone or the end of a traffic queue. The techniques and options for improving overall queue management may include:

- reducing queue length by reducing stoppage times within work zones or where Traffic Control Persons are positioned
- reducing the length of a single lane alternating traffic section—the distance between stop bars or TCPs
- scheduling the work activity at a time when traffic volumes are lower
- using additional signage in advance of the main construction zone signage[1] (see *Figure 6.3: Managing Queue Length and Additional Advance Warning*)
- undertaking frequent sign checks to ensure that all signs are erect and visible to approaching drivers
- adjusting the placement of signage so that it is in the most visible location, and repeating the signage at the approach to the work zone

Also consider any other conditions that may make a standard advance warning setup insufficient for the situation, including:

- weather conditions, such as fog, that make it difficult to see signs in the approach to a work area
- high winds or routes with limited or no shoulder space, making it difficult to erect signs with appropriate spacing and ensure that they stay erect

When such conditions exist, additional advance warning signage and sign checks may be required.

1 Use the Traffic Control Person Ahead C-001-1 sign only in the vicinity of a TCP, not in any other advance warning series.

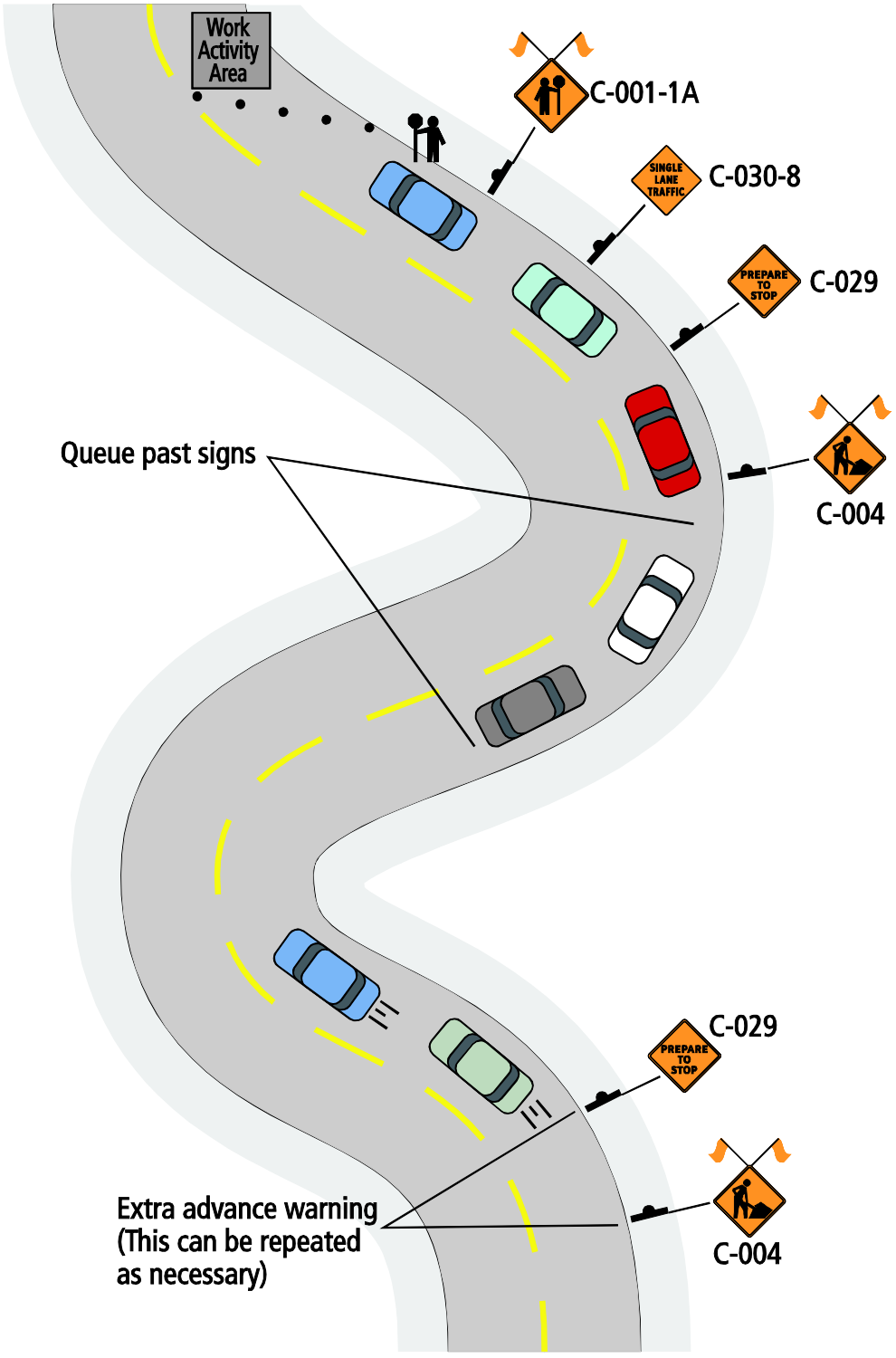



**Figure 6.3: Managing Queue Length and Additional Advance Warning**

### 6.4.2 Preventing Premature Queue Shifting

Premature queue shifting occurs when drivers move into the directed lane well upstream of the traffic control person. This creates problems for the following reasons:

- All vehicles in the queue must be released in order to clear the lane for opposing or work traffic.
- It is difficult to stop the queue for emergency or work functions.
- It is difficult to accommodate the passage of emergency vehicles.
- If the traffic is being controlled by traffic signals, the vehicles may no longer be passing over the detectors.

Using centreline delineation on the approach to a stop condition defines the path for traffic to follow and keeps vehicles in the appropriate lane until drivers are directed to shift. This delineation method is useful because:

- It keeps drivers in the required travel lane as they approach and depart from the Traffic Control Person.
- It provides visual cues to drivers that may assist in slowing the speed at which they join the queue.

See also *Figure 6.4: Premature Queue Shifting and Prevention of Premature Shifting.*

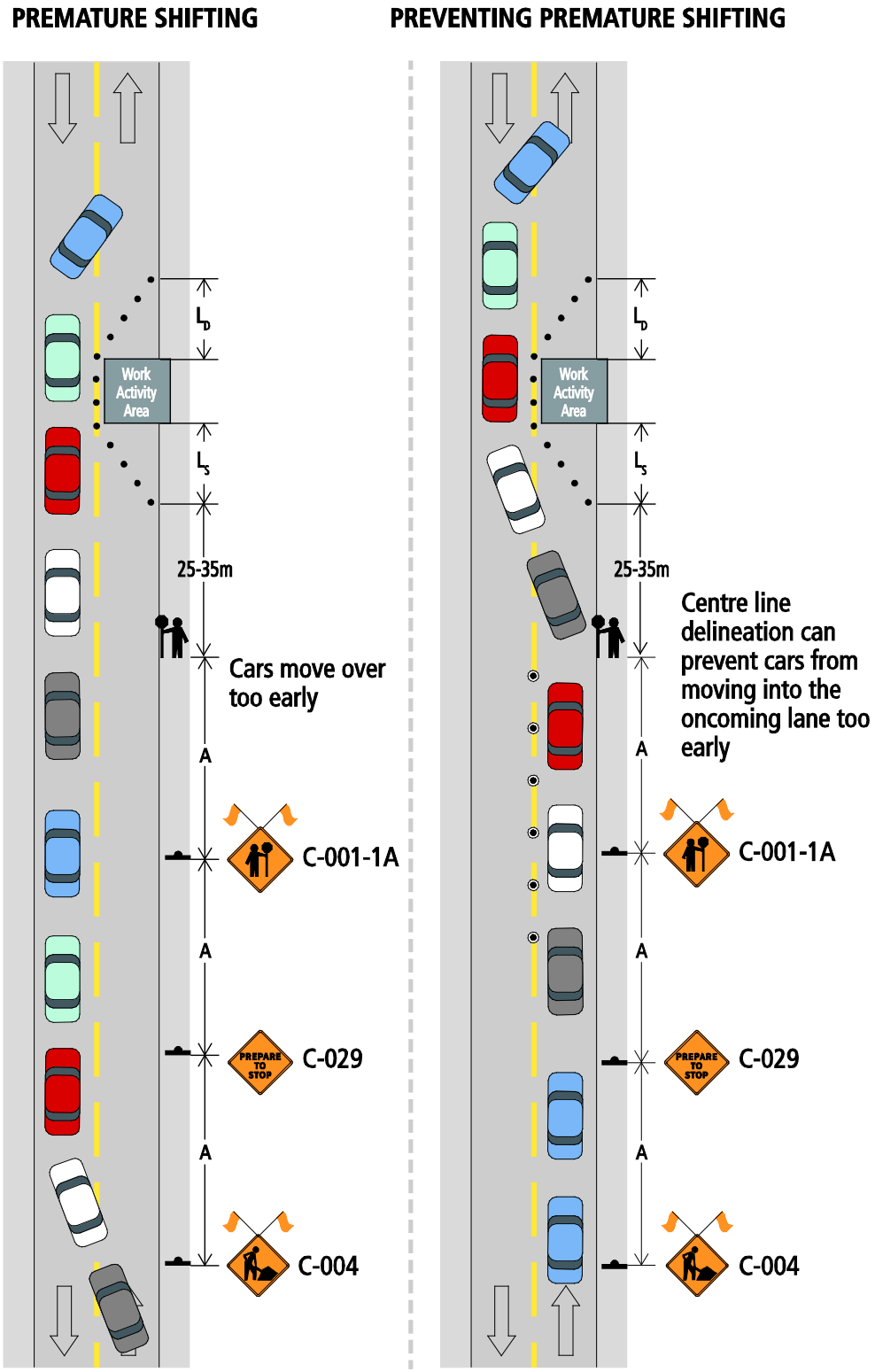



**Figure 6.4: Premature Queue Shifting and Prevention of Premature Shifting**

## 6.5 Treatment of Drop-Offs and Travel Lane Excavations

### 6.5.1 Drop-Offs

| | |
|---|---|
| **Drop-Off** | An abrupt change in elevation created by construction activity (such as milling, paving, or excavation) that is steeper than 3:1 (non-traversable slope). It is typically adjacent to a travel lane, and runs parallel to the direction of travel. |

Drop-offs in work zones should be marked with appropriate signing and devices.

Roadways with drop-offs should have regular, scheduled inspections—including over weekends and holidays—to ensure that the appropriate devices are in place.

Drop-offs that are left exposed to traffic shall be treated as follows:

1. If the drop-off is caused by the removal of barrier, Barrier Removed C-069 signs shall be installed in advance of the drop-off and every 500 metres as long as the condition persists.
2. Drop-offs ≤ 60 mm (2.5”) should be signed with a Low Shoulder C-013 sign to alert motorists to the condition. The sign should be repeated at least every 500 metres as long as the condition persists.

   On two-lane, two-way roadways, the sign may have to be applied for both directions of travel, and “no passing” restrictions may have to be considered. This type of drop-off is common in shoulder rehabilitation and gravel shoulder maintenance projects.
3. Drop-offs > 60 mm (2.5”) but ≤ 130 mm (5”) should be delineated with tubular markers or drums, and signed with Low Shoulder C-013 signage. The tubes or drums should be spaced according to *Table B – Device Spacing Lengths* (see *Appendix F*).
4. For drop-offs > 130 mm (5”) but < 300 mm (12”), a speed reduction to 80 km/h or less is required, as well as appropriate signage and delineation using tubes or drums.
5. The following treatment options exist for drop-offs ≥ 300 mm (12”):
   - Where the drop-off will be present for 48 hours or less, the drop-off treatment may be as indicated in #3 above; or
   - Closure of an adjacent travel lane in the same direction (for multi-lane highways); or
   - Creation of a lane shift to move traffic away from the drop-off by at least the distance shown in *Table 6.1: Minimum Distance from Edge of Travel Lane to Drop-Off*, in addition to the treatment indicated in #3 above; or
   - Installation of a temporary barrier with at least 300 mm (12”) between the back of the barrier and the drop-off, and at least 600 mm (24”) between the front of the barrier and the edge of the travel lane. An approved barrier terminal, flare, or crash attenuator is required at the beginning of the barrier section, and the barrier should have retroreflective markers and/or warning lights for night-time use.

When considering the options above, reducing the lane width to 3.2 metres is generally acceptable.

For situations in which no temporary barrier is installed, *Table 6.1: Minimum Distance from Edge of Travel Lane to Drop-Off* shows the minimum distance required between the edge of a travel lane (as marked by a fog/edge line or a line of channelizing devices) and a drop-off ≥ 300 mm (12”).

**Table 6.1: Minimum Distance from Edge of Travel Lane to Drop-Off**

| Regular (Non-Construction) Speed Limit (km/h) | Distance from Edge of Travel Lane to Drop-Off (m) |
|---|---|
| ≤ 60 | 0.5 |
| 70 - 100 | 1.5 |
| ≥ 110 | 3.0 |

Example:

On a 110 km/h highway, a 60 cm (24”) deep excavation is required alongside the travel lane. The project team determines that if they apply a lane shift to move traffic away, traffic will be shifted by only 1.8 metres.

Since a 3-metre offset is the minimum required for a 110 km/h highway, the lane shift is not an acceptable option. Instead, the project team should consider closing a lane or installing a temporary barrier.

If a highway design Engineer prepares a design that deviates from the above guidelines, the design shall include a written, stamped, and documented decision that provides the rationale for the alternative strategy.

Possible reasons for using an alternative strategy may include very low traffic volumes, very short work duration, or mitigation of risk through the use of enhanced signing or enhanced enforcement.

### 6.5.2 Travel Lane Excavations

| | |
|---|---|
| **Excavation** | An abrupt change in elevation created by construction activity within the travelled portion of the roadway. |

Excavations ≥ 60 mm (3”) that are left unattended and exposed to traffic shall be treated as follows:

1. If the regular (non-construction) speed limit is ≥ 70 km/h, excavations may be treated by:
   - Backfilling the excavation, paving to match the existing grade, and installing a Bump or Rough Roadway Ahead C-017 sign ahead of the filled excavation if there is any noticeable difference in elevation; or
   - For situations that are present for no more than 72 hours, backfilling the excavation with gravel and compacting it to match the existing road grade, without paving. A Motorcycle Rough Surface C-019 sign with a Gravel Surface C-019-T tab shall be installed ahead of the filled excavation, and the surface should be monitored and maintained as a smooth surface, free of rutting and wash boarding.

2. If the regular (non-construction) speed limit is ≤ 60 km/h, excavations may be treated by:
   - Installing a steel plate across the excavation and installing a Bump or Rough Roadway Ahead C-017 sign or another appropriate warning sign ahead of the covered excavation; or
   - Backfilling the excavation with gravel and compacting it to match the existing road grade. A Motorcycle Rough Surface C-019 sign with a Gravel Surface C-019-T tab shall be installed ahead of the filled excavation.

If it is not feasible to cover or fill the excavation, the affected area will have to be closed to traffic. This may require a full or partial lane closure, with additional traffic control.

## 6.6 Positioning of Temporary Traffic Control Devices

Roadway tapers are important components of temporary work zones. They are created by using a series of channelizing devices placed to move traffic out of or into its normal path. Table A shows the taper lengths appropriate for various speed limits.

Device spacing on entry to a work zone should be based on the regular posted speed limit of the highway, regardless if a reduced speed is implemented prior to the work zone. The reasons for this are:

1. The 2015 update to the 1995/1999 Traffic Control Manual for Work on Roadways was in part driven by the need to address the Ministry's increased use of night work and subsequent traffic control for highway maintenance and construction.
2. Nighttime brings a reduction in visibility for drivers, and drivers are often less alert during this time period.
3. To mitigate nighttime driver issues of reduced alertness and slower response times, providing greater distances for warning signs and taper transitions may reduce conflicts at merge and shift points.

Options for lengthy work zones where additional work areas may be utilized, including work zones with multiple construction speed limits:

1. All additional signing, tapers, and device spacing within the work zone may be deployed using the regular posted speed limit of the highway.

   *Example: Regular posted speed limit of highway =120 km/h, and construction speed limit reduced to 80 km/h. All signing, tapers and devices deployed continue to use the 120 km/h speed.*

2. Additional signing, tapers, and device spacing for additional work areas may be based (if a speed reduction is utilized) on the new reduced construction speed limit if signing, tapers, and devices deployed **on entry** have been spaced to the regular posted speed limit, or

   *Example: Regular posted speed limit of highway =120 km/h, all signing, tapers and devices deployed **on entry** use this 120 km/h speed. Reduced construction speed limit is now 80 km/h where additional signing, tapers, and devices may use the 80 km/h speed for layout.*

3. If another additional reduced speed limit is deployed beyond the first construction speed reduction (if a speed reduction was utilized) the signing, tapers, and device spacing for this new reduced construction speed zone will be based on the speed prior to entry to the new speed reduced zone.

   *Example: Regular posted speed limit of highway =120 km/h, all signing, tapers and devices deployed on entry use this 120 km/h speed. First reduced construction speed limit is to 80 km/h, where additional work areas may utilize options 1 or 2. However, an additional work area now calls for a further reduced construction speed limit from the 80 km/h to 60 km/h. In this case, additional signing, tapers, and devices **on entry** to the 60 km/h speed zone may use either:*

   *a) the regular posted speed limit of 120 km/h, or*

   *b) utilize the reduced construction speed limit of 80 km/h for the layout **on entry** to the 60 km/h area.*

A minimum of five devices are required for any taper.

| TABLE A – TAPER LENGTHS | | | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| **Taper Types (m)** | | **Regular Posted Speed Limit (km/h)** | | | | | | | |
| | | **≤50** | **60** | **70** | **80** | **90** | **100** | **110** | **120** |
| **Merge Taper Length** | **$L_M$** | 35 | 55 | 160 | 190 | 210 | 230 | 250 | 280 |
| **Lane Shift Taper Length** | **$L_L$** | 30 | 50 | 80 | 100 | 110 | 120 | 130 | 140 |
| **Downstream Taper Length** | **$L_D$** | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 |
| **TCP, Signal, and Shoulder Taper Length (min. 5 devices)** | **$L_S$** | 5 | 8 | 15 | 15 | 15 | 15 | 15 | 15 |
| **Minimum Tangent Length between Tapers** | **$L_T$** | 30 | 60 | 160 | 190 | 210 | 230 | 250 | 280 |
| **Run-In Length on Centreline** | **$L_R$** | 40 | 50 | 60 | 60 | 70 | 80 | 90 | 100 |

| Table A Notes | |
|---|---|
| **Regular Posted Speed Limit** | Device spacing and taper lengths should be to the regular posted speed limit. |
| **$L_M$ = Merge Taper Length** | Merge length required to close lane on approach to work area. For speeds ≥ 70 km/h, merge length should be at least = $\frac{(\text{lane width of } 3.7 \text{ m}) \times (\text{Posted Speed in km/h})}{1.6}$, rounded to nearest 10 m. |
| **$L_L$ = Lane Shift Taper Length** | Used when a lateral shift is needed within the work area.<br>Lane Shift Taper = ½ x $L_M$, rounded up to nearest 10 m. |
| **$L_D$ = Downstream Taper Length** | May be used in work zone termination area to provide a visual cue to drivers that they may return to the original lane or path that was closed. |
| **$L_S$ = TCP, Signal, and Shoulder Taper Length** | Shoulder Taper: Used to close shoulders within activity area, or when shoulders might be mistaken for driving lanes. May be increased to 1/3 x $L_M$ on higher-speed highways and freeways where shoulder width is ≥ 2.5 m.<br>Signal and TCP Tapers: Used in advance of a work activity area where traffic is controlled so that the road is used alternately by traffic moving in each direction. |

| | |
|---|---|
| **$L_T$ = Minimum Tangent Length between Tapers** | Used between successive tapers or at other decision or conflict points to provide time for drivers to become accustomed to the first change and observe traffic control devices for the second change.<br>**$L_T$ = $L_M$**, but for high-speed/high-volume freeways and/or night work, it may be doubled (2 x $L_T$) to increase time for drivers to become accustomed to the first change. |
| **$L_R$ = Run-In Length on Centreline** | May be used on centrelines as minimum tangent length before development of lane departures or lane shifts.<br>Run-in length = 0.8 x speed (in km/h) (US Manual of Uniform Traffic Control Devices). |

*Figure 6.5: Taper Diagram* shows a general concept of the lengths found in Table A. For traffic control device selection, refer to the appropriate reference layout found in Sections 7 to 19.

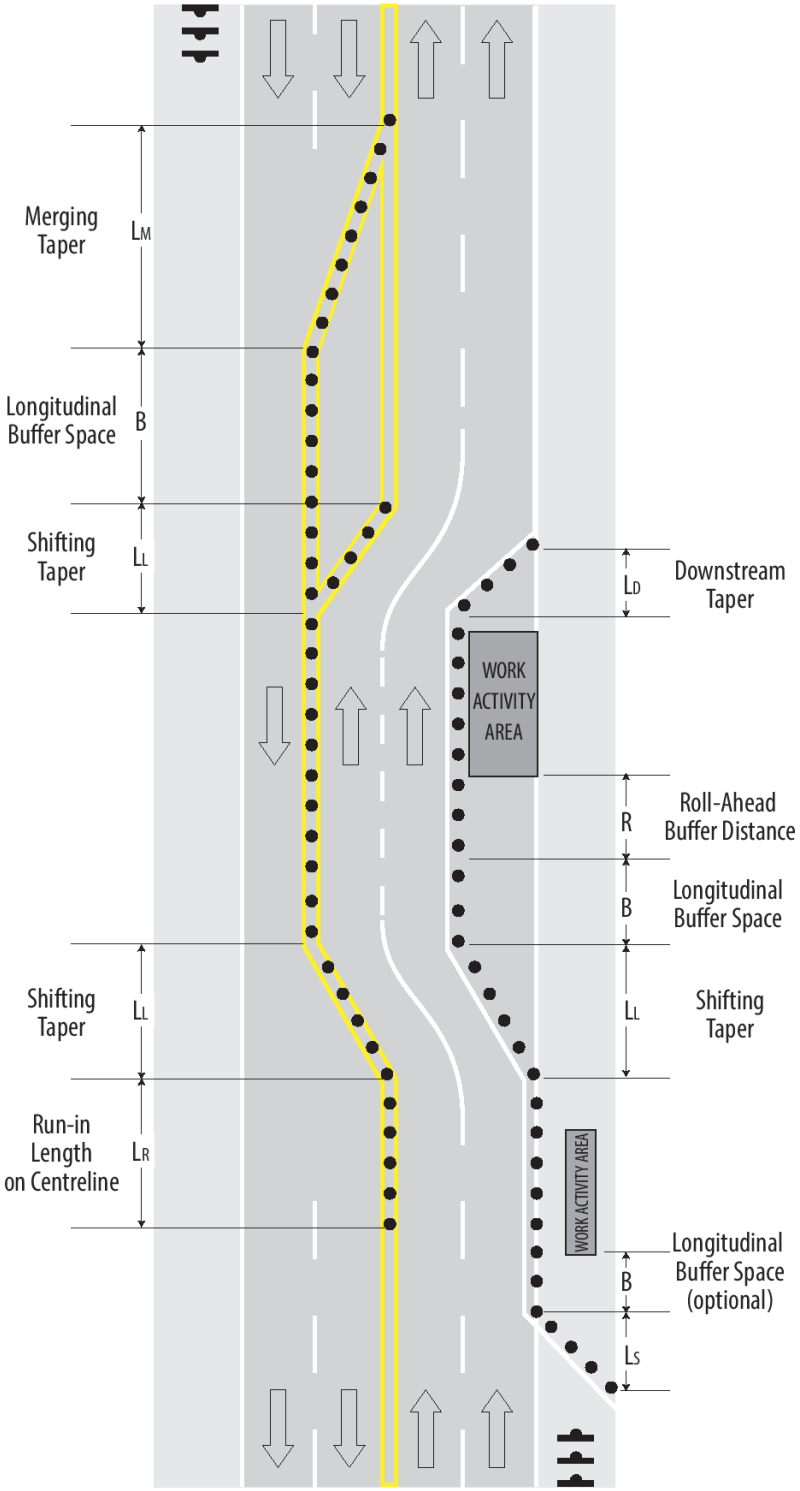



**Figure 6.5: Taper Diagram**

Tapers require at least five devices—typically drums or tubes.

To set the taper:

1. Place Device 1 as noted in *Figure 6.6: Taper Layout.*
2. Measure the taper length (parallel to the roadway).
3. Place Device 2 as noted in the figure.
4. Fill in the area between the two devices with devices of the same type, using the spacing shown in *Table B – Device Spacing Lengths* as the maximum distance between devices.
5. Adjust the device spacing as necessary.

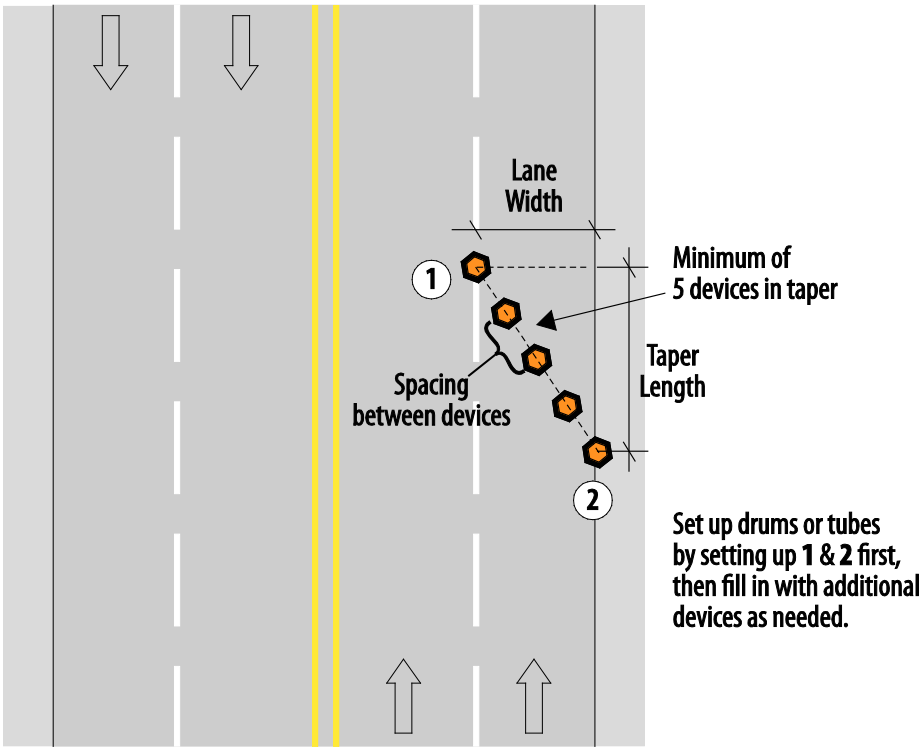



**Figure 6.6: Taper Layout**

The appropriate positioning of traffic control devices varies with the regulatory speed limit for the roadway and the traffic control devices being used. Some distances shown for 50 km/h and 60 km/h speed limits in Table B are from the BC Traffic Control Manual for Work on Roadways (1999).

**TABLE B – DEVICE SPACING LENGTHS**

| Device Spacing (m) | | Regular Posted Speed Limit (km/h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | ≤50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 |
| **Construction Sign Spacing** | **A** | 40 | 60 | 80 | 100 | 150 | 150 | 200 | 200 |
| **Buffer Space** | **B** | 30 | 40 | 60 | 80 | 110 | 140 | 170 | 200 |
| **Roll-Ahead Buffer Distance** | **R** | 30 | 30 | 40 | 40 | 40 | 50 | 50 | 50 |
| **Channelizing Device Spacing for Tapers** | **C** | 10 | 10 | 15 | 15 | 15 | 15 | 15 | 15 |
| **Channelizing Device Spacing on Curves and Tangents** | **D** | 10 | 10 | 30 | 30 | 40 | 40 | 40 | 50 |

| Table B Notes | |
|---|---|
| **Regular Posted Speed Limit** | Device spacing and taper lengths should be to the regular posted speed limit. |
| **A = Construction Sign Spacing** | Recommended minimum spacing for signage. Spacing may be adjusted to accommodate site constraints and/or where high numbers of access points exist. Signs within the work zone should be spaced on the basis of the pre-construction, regulatory speed limit.<br>Maximum Construction Sign Spacing:<br>• Spacing for the sign closest to the work activity area should remain as close as possible to Distance A.<br>• For other construction signs in the advance warning area, spacing may be adjusted up to a maximum distance of 2 x Distance A.<br>• Signs that include a distance measurement (e.g., Construction Ahead Next 2 km) should be placed in accordance with the distance cited on the sign or tab. |

| | |
|---|---|
| **B = Buffer Space** | The longitudinal distance which provides a margin of safety for both the driver and the workers. It is important that the buffer space be free of equipment, workers, material and vehicles.<br><br>A buffer vehicle with a crash attenuator may be located within the buffer space if there are space constraints.<br><br>The buffer space is measured is from the end of the taper to the work activity area unless there is a buffer vehicle, in which case it is measured to the back of the buffer vehicle.<br><br>Typically used on high-speed roadways but should be considered for all works where space allows.<br><br>Distance is based on the braking distance on level ground for wet pavement as defined in the Transportation Association of Canada's Geometric Design Guide for Canadian Roads (1999). |
| **R = Roll-Ahead Buffer Distance** | The longitudinal distance measured from the front of the buffer vehicle to the work activity area. It provides a margin of safety in case of impact.<br><br>Distance is based on the Minnesota Manual on Uniform Traffic Control Devices (2011). |
| **C = Channelizing Device Spacing for Tapers** | Maximum spacing between channelizing devices for tapers. |
| **D = Channelizing Device Spacing on Curves and Tangents** | Maximum spacing between channelizing devices on curves and tangents. Tighter spacing is acceptable especially on curves where device loss can impact directional continuity.<br><br>Maximum device spacing is calculated as 0.4 x speed (in km/h), rounded to the nearest 10 m. |

## 6.7 Device Installation and Removal

Device installation and removal must be carried out only by individuals who have suitable traffic management work experience or training. If flagging is required during installation and removal, then TCPs will need to be present to direct traffic.

Below is the Ministry's preferred approach for installing and removing temporary traffic control devices. The goal is to ensure that devices are installed efficiently, safely, and in the correct location with the correct spacing. There may be other methods which will also achieve this goal.

### 6.7.1 General Guidelines

1. It is important to develop the Traffic Control Plan before setting up signs and devices so that the appropriate distance is implemented between the work zone and the traffic control sign or device furthest upstream.
2. Flashing arrow boards, Traffic Control Persons, and/or flashing vehicle lights should be used during device installation and removal as required. Drivers may not expect to encounter workers on the roadway until devices are installed (see *Section 4.6: Flashing Arrow Boards (FABs)*).
3. More than one work vehicle may be used to install and remove devices.
4. When signs or channelizing devices are to be installed and removed several times during the work operation, the device locations may be marked so that the installation can be repeated quickly and with proper placement assured.
5. When not required, the devices should be stored off the roadway or out of sight.
6. Traffic Control Person Ahead C-001 signs shall remain in place until Traffic Control Persons are not actively controlling traffic.
7. Travelling against the flow of traffic is only permitted within a closed lane.
8. When setting up and removing devices, there is typically no speed reduction so the original posted speed limit of the roadway determines the need for shadow vehicles and crash attenuators.

For further guidance on the use of shadow vehicles when conducting traffic control device placement and/or removal, see *Section 10.3: Intermittently-Moving Work – Two-Lane, Two-Way Roadway* and *Section 10.5: Intermittently-Moving Work – Multilane Undivided or Divided Roadway*.

### 6.7.2 Installation Considerations

1. If traffic speed is ≥ 70 km/h, a shadow vehicle with a 360-degree flashing light and 4-way flashers is recommended when installing traffic control devices. Position the shadow vehicle between the worker and the approaching traffic.
2. If the signs are not required immediately, turn them so that they are not visible to traffic until they are needed.
3. Cover any existing or conflicting signs on the roadway when the temporary signs are in effect.
4. A Traffic Control Person may be required for stopping traffic while channelizing devices are being placed around the work activity.
5. Ensure that signs and channelizing devices are visible to oncoming traffic. Adjust the signs as required.
6. Work may commence once all temporary traffic control devices are in place.

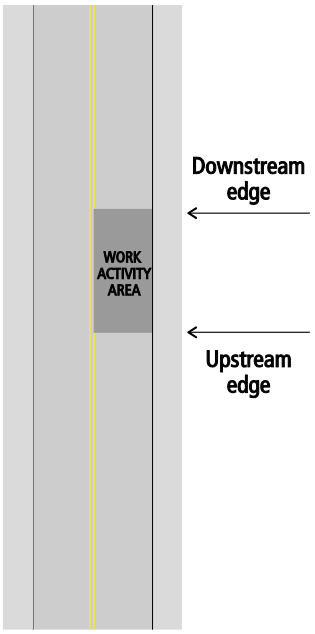
### 6.7.3 Removal Considerations

1. If the speed is ≥ 70 km/h, a shadow vehicle with a 360-degree flashing light and 4-way flashers is recommended when removing traffic control devices. Position the shadow vehicle between the worker and the approaching traffic.
2. Uncover any previously covered signs that are required for the two-way traffic.
3. In detour situations where vehicles have been directed to use alternative routes, remove upstream signage first to ensure that traffic has not been directed to an alternative route from which the signs have been partially removed.

### 6.7.4 Two-Lane, Two-Way Roadways

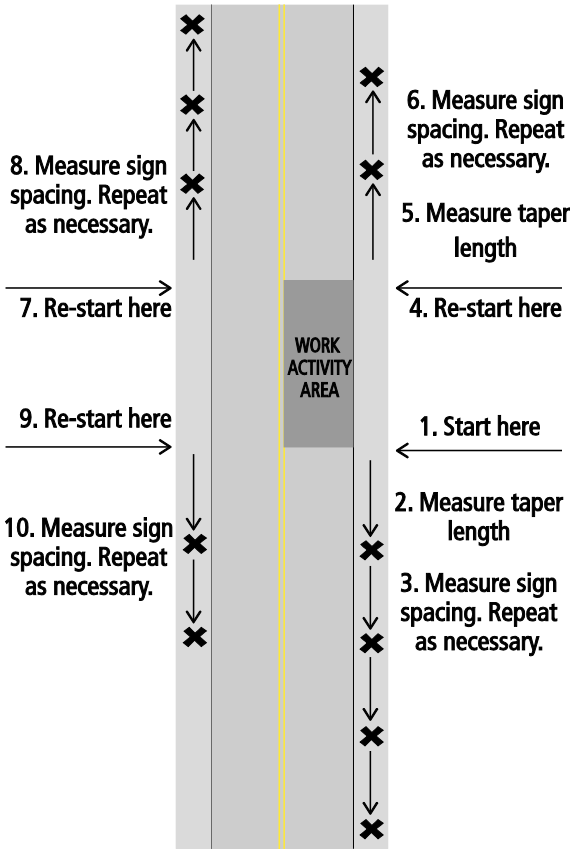
**Step 1: Planning the Setup**

1. Using the Traffic Control Plan, define the edges of the work activity area.
2. Mark the upstream and downstream edge of the work activity area.




**Step 2: Determining the Layout**

1. Mark sign and channelizing device placement in accordance with the Traffic Control Plan or Tables A and B (see *Section 6.6* or *Appendix F*).
   - Measurements may be made using a variety of methods (e.g., Distance Measuring Instrument, GPS, or manual measuring).
   - Marking may be done at the edge of pavement with survey stakes, paint, or other temporary markers.
   - Measure outward starting from the edge of the work activity area.
2. Follow sub-steps 1 through 10 as shown in the diagram on the right.

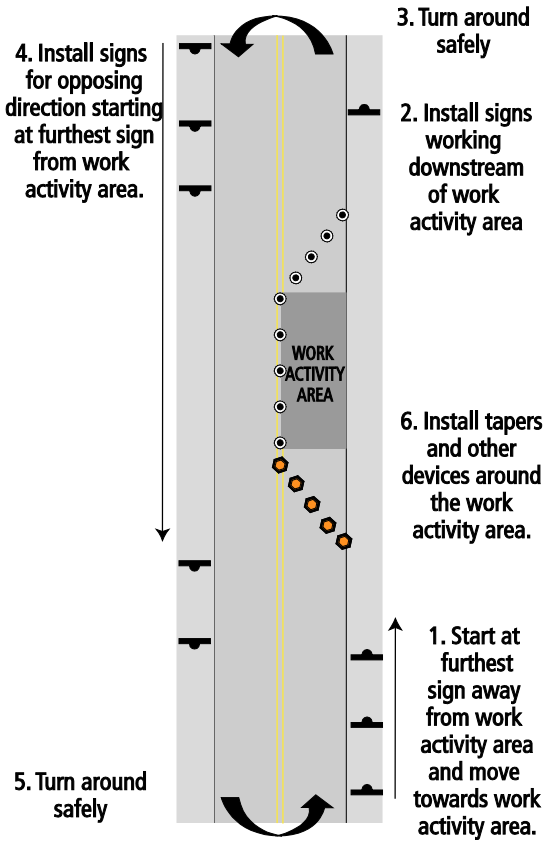


### Step 3: Order of Installation

Traffic control devices should be placed in the order that drivers will encounter them, beginning with the sign or device furthest upstream from the work activity area and continuing forward. Typically, channelizing devices forming lane shifts, merges, detours, and other traffic pattern changes are established after the signs identifying the work zone are in place.

Using the diagram on the right:

1. Start at the furthest sign upstream and move towards the work activity area, placing signs in the locations previously marked and keeping as far to the right as possible.
    - Cover any existing conflicting signs.
2. Place signs for the termination area.
3. Pull off the road in an appropriate area to turn around and travel in the opposite direction.
4. Complete sign placement in the opposite direction, keeping as far to the right as possible.
    - Cover any existing conflicting signs.
5. Pull off the road in an appropriate area to turn around and travel in the opposite direction.
6. Place tapers and delineation around the work activity area.




### Step 4: Planning the Removal

As soon as the traffic control devices are no longer needed, they should be removed. During the removal process, Traffic Control Persons, flashing arrow boards, shadow vehicles, and/or flashing vehicle lights should be used.

Signs and devices are removed in the opposite order of which they were installed. Traffic Control Person Ahead C-001 signs should remain in place until Traffic Control Persons are no longer needed.

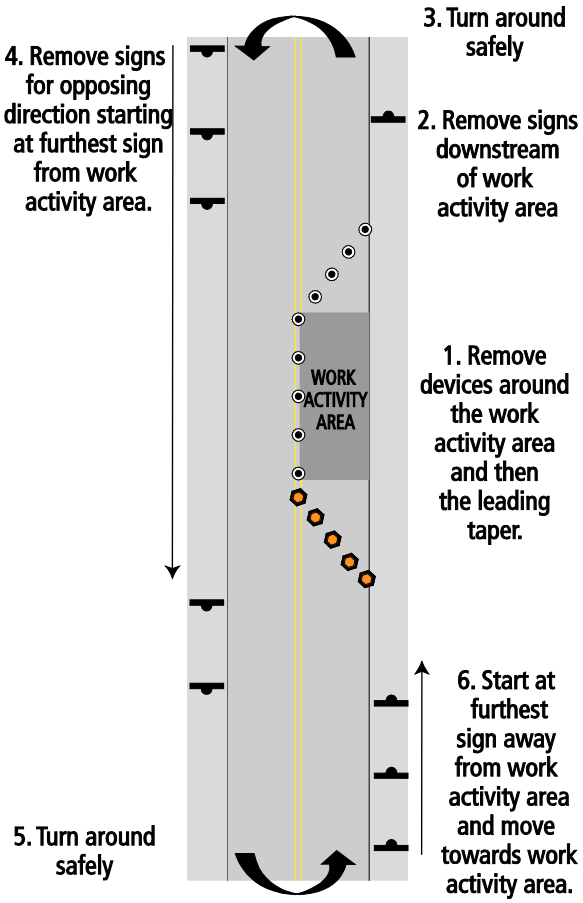
No worker should ride outside on the rear of a reversing vehicle. If special circumstances allow for this practice, it is mandatory to follow Section 16.31 in WorkSafeBC's Occupational Health and Safety Regulation (Rider Restriction).

### Step 5: Order of Removal

Ensure that work operations have ceased, and that all equipment and workers are off the roadway.

Using the diagram on the right:

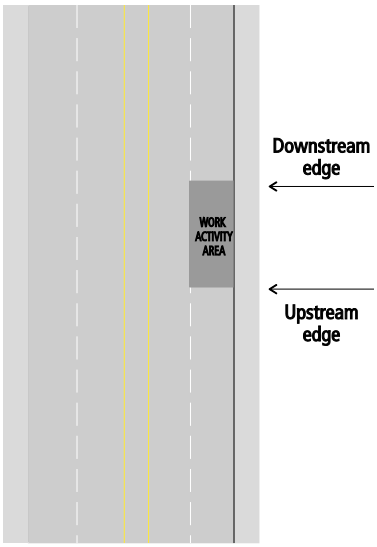
1. First remove channelizing devices on the travelled roadway. Do not remove advance warning signs until all other devices are removed.
    - Use a TCP to stop traffic, if required, and pull channelizing devices off the roadway, starting with those closest to the work on the downstream side and then moving to the upstream side. If necessary, place channelizing devices on the shoulder for storage and later pick-up.
2. Starting at the nearest sign downstream of the work activity area, remove or turn signs not in use, keeping as far to the right as possible.
    - Uncover any previously covered signs that are required for re-establishing normal traffic operations.
3. Pull off the road in an appropriate area to turn around and travel in the opposite direction.
4. Remove devices and remove or turn signs not in use in the opposite direction, keeping as far to the right as possible.
    - Uncover any previously covered signs that are required for re-establishing normal traffic operations.
5. Pull off the road in an appropriate area to turn around and travel in the opposite direction.
6. Remove the advance warning signs upstream of the work activity area.



### 6.7.5 Multilane Roadways

#### Step 1: Planning the Setup

1. Using the Traffic Control Plan, define the edges of the work activity area.
2. Mark the upstream and downstream edge of the work activity area.




#### Step 2: Determining the Layout

1. Mark sign and channelizing device placement in accordance with the Traffic Control Plan or Tables A and B (see *Section 6.6* or *Appendix F*).
    - Measurements may be made using a variety of methods (e.g., Distance Measuring Instrument, GPS, or manual measuring).
    - Measurements may be made on one side of the road for signs on both sides of the roadway.
    - Marking may be done at the edge of pavement with survey stakes, paint, or other temporary markers.
2. Follow sub-steps 1 through 10 as shown in the diagram on the right.

